
Chapter 11: Expressions and Equations

Worksheet 11: Linear Equations in One Variable

Learning Objective: Students will solve linear equations in one variable (with variables on both sides, with fractions, and in word problems); verify solutions; and apply equations to real-life problems.

Worked Example

Solve: $\frac{2x+3}{3} - \frac{x-1}{2} = 1$

Solution: Multiply both sides by $\text{LCM}(3,2) = 6$:

$$2(2x+3) - 3(x-1) = 6$$

$$4x + 6 - 3x + 3 = 6 \Rightarrow x = -3$$

Check: $\frac{-3}{3} - \frac{-4}{2} = -1 + 2 = 1 \checkmark$

Questions 1–5: Easy / Recall

Q1. Solve: $3x + 7 = 22$

Q2. Solve: $5y - 3 = 2y + 9$

Q3. Solve: $\frac{x}{4} = 7$

Q4. Solve: $2(x - 3) = 8$

Q5. If $4x - 5 = 19$, find $3x + 2$.

Questions 6–10: Basic Application

Q6. Solve: $\frac{3x - 1}{4} = \frac{x + 2}{3}$

Q7. The sum of three consecutive integers is 78. Find them.

Q8. A number is increased by 25% to give 75. Find the original number.

Q9. Solve: $7(x - 2) - 3(x + 1) = 5x - 11$

Q10. A father is 3 times as old as his son. In 12 years, he will be twice as old. Find their present ages.

Questions 11–15: Practice and Skill Building

Q11. Solve: $\frac{x + 1}{3} + \frac{x - 1}{4} = \frac{2x + 1}{6}$

Q12. The perimeter of a rectangle is 64 cm. Its length is 4 cm more than twice its breadth. Find the dimensions.

Q13. Solve: $0.4x + 0.3(x - 1) = 0.2(x + 2)$

Q14. The digits of a 2-digit number sum to 10. If the digits are reversed, the number increases by 36. Find the original number.

Q15. A train covers a distance at 60 km/h. If speed is increased by 15 km/h, it saves 2 hours. Find the distance.

Questions 16–18: Higher Order Thinking

Q16. The ratio of the present ages of two brothers is 4 : 3. After 6 years, the ratio will be 6 : 5. Find their present ages.

Q17. A cistern can be filled by pipe A in 4 hours and by pipe B in 6 hours. Pipe C can empty it in 8 hours. If all three are opened simultaneously, how long to fill the cistern?

Q18. Solve: $\frac{4x - 3}{3x + 2} = \frac{2x + 5}{3x - 1}$ and verify your answer.

Questions 19–20: Real-Life Applications

Q19. Meena spent $\frac{1}{3}$ of her pocket money on books, $\frac{1}{4}$ on stationery, and saved the remaining Rs. 150. Find her total pocket money. (Set up and solve a linear equation.)

- Q20.** A shopkeeper has 25-paise and 50-paise coins totalling Rs. 45. The number of 25-paise coins is 30 more than the 50-paise coins. Find the number of each type of coin.

Reflection Question: An equation has the variable on both sides. What strategy helps you solve it efficiently? Demonstrate with an example of your choice.